

7	(a)	(i)	displacement antinodes at where there are no heaps correct shape	
		(ii)	$2.5\lambda = 39 \text{ cm}$	
		(iii)	$v = f\lambda$ $v = 2.14 \times 10^3 \times 15.6 \times 10^{-2}$ $= 334 \text{ m s}^{-1}$ A1	
		(iv)	length of tube $= 3.5 \times 15.6 \times 10^{-2} = 54.6 \text{ cm}$ $L = \frac{1}{2} \lambda$ so $\lambda = 109.2 \text{ cm}$ $f = v/\lambda = 305.7 = 306 \text{ Hz}$ (3 s.f.)	
		(v)	clearly defined heaps of dust are no longer observed fundamental mode for a closed pipe of 54.6 cm requires 153 Hz (OR the fundamental frequency required would be different)	
	(b)	(i)	$\pi \text{ rad}$	
		(ii)	1.5 wavelengths $= 18.6 - 14.4 = 4.2$ (length across 4 points) wavelength $= 2.8 \text{ cm}$ speed $= 2.8 \times 10^8 \text{ m s}^{-1}$	
		(iii)	metal plate moving away at 1.0 cm s^{-1} for every half-wavelength moved away, a high intensity signal is registered $f = 1.0 / (\frac{1}{2} \times 2.8) = 0.71 \text{ Hz}$	
		(iv)	intensity $\propto \text{amplitude}^2$ 0.25 ($\frac{1}{4}$ not accepted)	
		(v)	$\frac{1}{4} A + \frac{1}{4} A = \frac{1}{2} A$ so maximum intensity $= 0.25 \times \text{original intensity}$ max is now less than original, 0.25 <i>I</i> no change in positions of nodes or antinodes	

8	(a)	the e m f induced in a conductor is directly proportional to the	
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